

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

EE1001-2 – BASIC ELECTRICAL ENGINEERING
 (AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

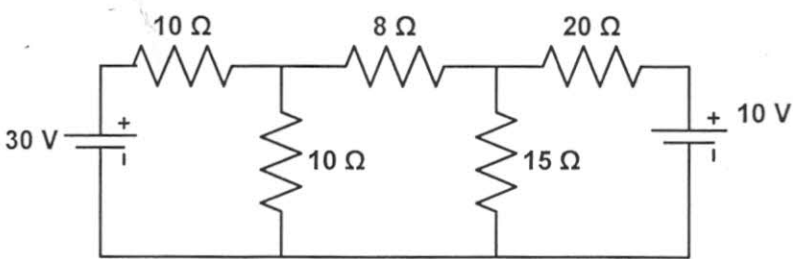
PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Peak factor of a sinusoidal current is _____**
 A) 0.707 B) 1.414
 C) 1.11 D) None of these
2. **If an alternating current has its RMS value 5 A, frequency 60 Hz, its instantaneous value is given by _____**
 A) $i = 5 \sin 377t$ A B) $i = 5/\sqrt{2} \sin 377t$ A
 C) $i = 5\sqrt{2} \sin 377t$ A D) None of these
3. **The power factor of an AC circuit is given by _____**
 A) Cosine of the phase angle between Voltage and current B) Tangent of the phase angle between Voltage and current
 C) The ratio of R by XL D) The ratio of XL by Z
4. **If a 100V, 50Hz, single phase AC supplies a current of 2A to a pure capacitive circuit, the capacitance of the circuit is _____**
 A) 50 F B) 637mF
 C) 63.7 μ F D) 0.159F
5. **The capacitive reactance is measured in _____**
 A) ohm B) farad
 C) henry D) None of these
6. **When a Pure resistive Circuit, is energized by an AC supply, the angle between voltage and its current is _____**
 A) 90° B) In between 0° and 90°
 C) 0° D) None of these
7. **Kirchhoff's laws are useful in determining _____**
 A) Current flowing in a circuit B) EMFs and Voltage drops in a circuit
 C) Power in a circuit D) All the above
8. **In a phasor diagram, the relationship between the voltage and its current when a pure capacitive circuit is energized by an AC supply is _____**
 A) Voltage lags its current by 90° B) Voltage leads its current by 90°
 C) Voltage lags its current by angle in between 0° and 90° D) Voltage leads its current by angle in between 0° and 90°
9. **The reluctance of a magnetic material _____**
 A) Decreases with increasing cross-sectional area of material B) Increases with the increasing cross-sectional area of material
 C) Does not vary with the increasing cross-sectional area of material D) None of these
10. **The material used for brushes of DC generator is _____**
 A) Carbon B) Copper
 C) Both (a) and (b) D) None of the above

11. In an induction motor, the difference between the synchronous speed of the magnetic field and the actual speed of the rotor is called as the
 - A) Synchronous speed
 - B) slip speed
 - C) Asynchronous speed
 - D) maximum speed
12. The number of parallel paths in 8 pole wave wound DC generator is
 - A) 2
 - B) 16
 - C) 8
 - D) 4
13. The secondary voltage of a 10kVA transformer with load current of 10A is _____
 - A) 10kV
 - B) 100kV
 - C) 1kV
 - D) none of the above
14. In a given transformer for given applied voltage, which of the following losses remain constant irrespective of load changes?
 - A) Friction and windage losses
 - B) Copper losses
 - C) Hysteresis and eddy current losses
 - D) Cannot be determined
15. A transformer is designed so that primary and secondary have _____
 - A) high leakage reactance
 - B) large resistance
 - C) tight magnetic coupling
 - D) good electrical coupling
16. A 2000/200V, 20kVA ideal transformer has 66 turns in the secondary. The number of primary turns is _____
 - A) 440
 - B) 660
 - C) 550
 - D) 330
17. The transmission unit in EV comprises of _____
 - A) Gear Box
 - B) Hinges and Levers
 - C) Rack and Pinion
 - D) Crank Shaft
18. The feature of Electric Motor important for an EV is _____
 - A) High Starting Torque
 - B) High power density
 - C) High efficiency
 - D) All of the above
19. In a boost converter which of the statement is true
 - A) Input voltage is more than output voltage
 - B) Input voltage is less than output voltage
 - C) Input voltage is same as output voltage
 - D) Input and output voltages are independent
20. Earthing is necessary to give protection against _____
 - A) voltage fluctuation
 - B) Overloading
 - C) electric shock
 - D) high temperature of the conductors

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define average value of an alternating quantity. Derive an expression for average value of an alternating current.	5	L*2	1	2
	b) Using mesh analysis, find the current through 15Ω resistor in the circuit shown in Fig: Q1b.				
 Fig: Q1b		7	L3	1	2
c)	In three-phase circuit two wattmeter used to measure power indicate 1200 W and 1800 W respectively. Find the power factor of the circuit.	4	L2	2	2
2.	a) With neat circuit diagram and waveforms derive the relationship between voltage and current in a single-phase RC circuit. Also derive the expression for power consumed.	5	L2	2	2

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|-------|--|---|----|---|---|
| b) | A coil of resistances $8\ \Omega$ and inductance 15mH is connected in series with a capacitor of capacitance $150\mu\text{F}$, across a single-phase supply of 230V , 50Hz . Calculate (i) impedance of the circuit (ii) current (iii) power consumed. | 7 | L3 | 2 | 2 |
| c) | Define the terms (i) phase (ii) power factor (iii) form factor (iv) frequency. | 4 | L1 | 1 | 2 |
| 3. a) | Discuss the principle of generation of alternating voltage. | 5 | L1 | 1 | 2 |
| b) | With a neat phasor representation obtain a relation between the line and phase voltages of a delta connected three phase balanced system. Obtain the expression for power. | 6 | L2 | 2 | 2 |
| c) | A balanced star connected load of resistance 8Ω and inductance 6Ω per phase is connected to a balanced 3-phase 400V supply. Find the line current, power factor and the power supplied to the load. | 5 | L3 | 2 | 2 |

Unit – II

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|-------|--|---|----|---|---|
| 4. a) | Derive the EMF equation of DC generator. | 5 | L2 | 3 | 2 |
| b) | Explain the principle of operation of a 3 Phase Induction Motor. | 6 | L2 | 3 | 2 |
| c) | In a 75 kVA , single phase transformer the iron and full load copper losses are 700W & 1200W . Find efficiency at 0.8 p.f lagging and load at which maximum efficiency occurs. | 5 | L1 | 4 | 2 |
| 5. a) | State and explain: | | | | |
| | (i) Faraday's Laws of Electromagnetic Induction | 5 | L2 | 4 | 2 |
| | (ii) Lenz Law. | 6 | L1 | 4 | 2 |
| b) | Explain different types of losses in transformer. | | | | |
| c) | A 230 V D.C. shunt motor takes 32 A at full load. Find the back e.m.f. on full load if the resistances of armature and shunt field windings are $0.2\ \Omega$ and $115\ \Omega$ respectively. | 5 | L3 | 3 | 2 |
| 6. a) | Explain the characteristics of DC series motor. | 5 | L2 | 3 | 2 |
| b) | A 125 kVA transformer has a primary voltage of 2000 volts at 50Hz . The number of primary turns are 182 and secondary turns are 40 . Neglecting losses, calculate (i) No load secondary e.m.f (ii) full load primary and secondary current. | 6 | L3 | 4 | 2 |
| c) | Explain the principle of operation single phase Induction motor. | 5 | L2 | 3 | 2 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | With neat diagram explain step up converter. | 8 | L3 | 5 | 2 |
| b) | With neat block diagram explain the different components used in Electric Vehicle. | 8 | L3 | 5 | 2 |
| 8. a) | Why earthing is essential and with neat block diagram explain pipe earthing. | 8 | L3 | 5 | 2 |
| b) | With neat diagram explain 3-way control of lamps. | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
